AI OPEN
DATA CENTER

Scaling Cooling for AI Factories - Latest Updates Technology Cooling Workstream

Dariusz Raczkowski Director, Client Services - Catalyst Commissioning Group LLC

Sara Hitt Technical Sales Manager, Whitespace - Vertiv

Jerrod Buterbaugh Principal - Thermal and Power Architecture - AMD

Rajat Bhagat - Senior Mechanical Engineer - Arcadis

Don Mitchell Technology Cooling Global Leader – Victaulic

OCP PLATINUM Member

Panel Discussion



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Senior Mechanical Engineer - Arcadis



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Technology Cooling Global Leader - *Victaulic*

OCP GLOBAL SUMMIT | 2026

How Much Does One Week Delay Cost



Assumptions

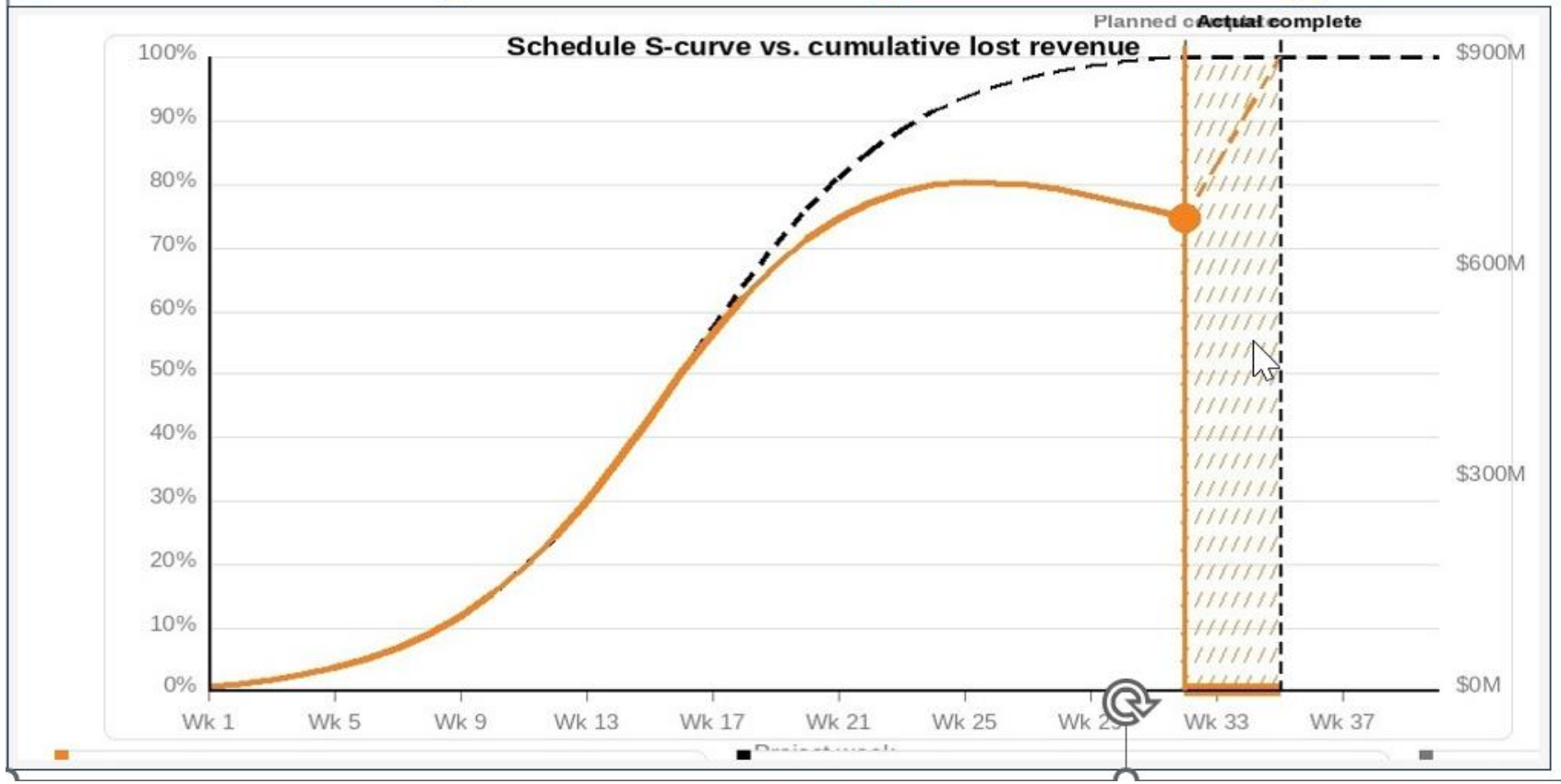
- Cost per rack: **$3.5 -5M**
- Revenue per rack: **$75M**
- Revenue realization: **3-4 years**
- Technology refresh cadence: **~2 years**
- Deployment size: **100 MW (600 racks)**

1 Week Delay "Time To Token"

$300 - $400 Million Lost Business

AI Factory – 3 Week Delay – Financial Impact

Planned Go-Live	Actual Go-Live	Delay	Lost Revenue	Daily Burn
Week 32	Week 35	3 Weeks	$900M	$42.9M



LOST REVENUE BY DELAY WEEK

Week +1

$300M (cumulative: $300M)

Week +2

$300M (cumulative: $600M)

Week +3

$300M (cumulative: $900M)

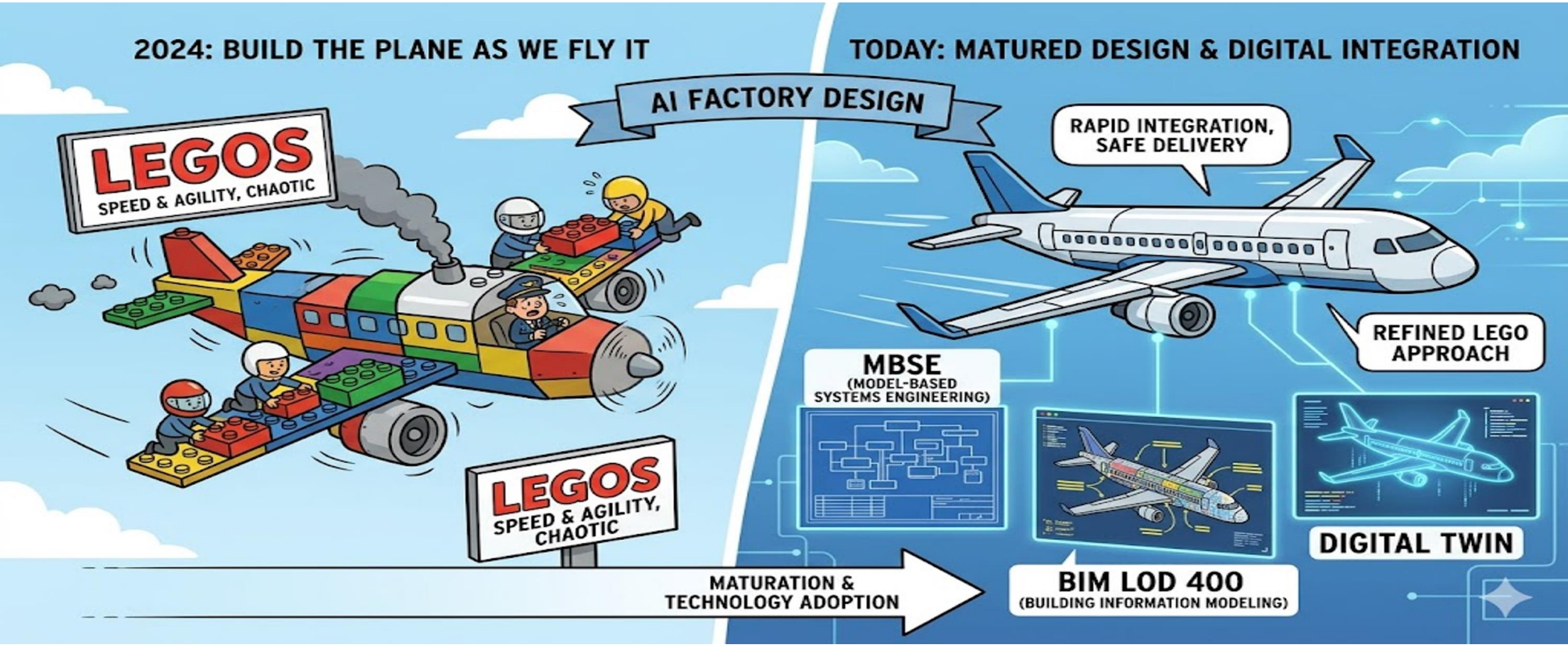
3-WEEK DELAY TOTAL

$900,000,000

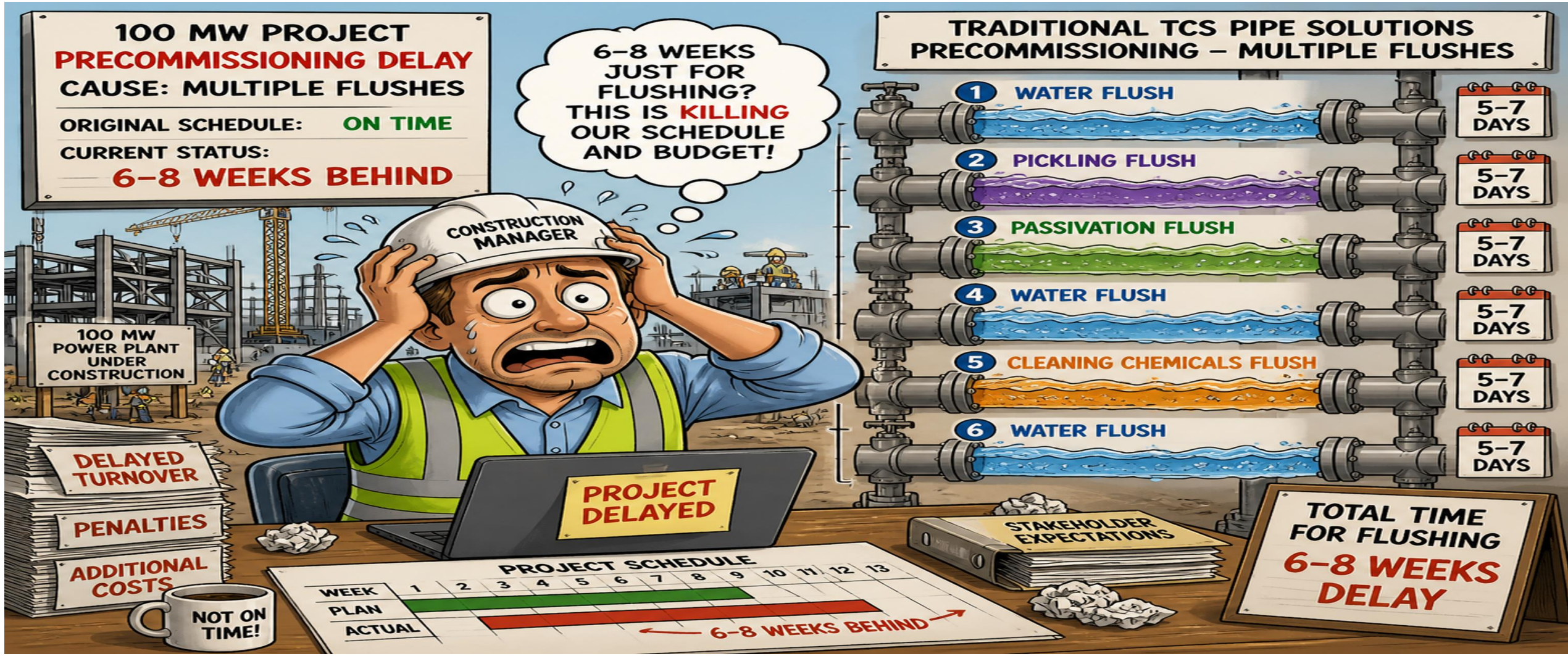
= 3 × $300M/week

Daily rate

AI Factory Digital Design, Delivery, Operation



Compress Pre-Commissioning: 7 Weeks => 1 Week



Compress Pre-Commissioning: Dusty, Not Dirty

THE GOAL:

All components pre-cleaned and assembled to a "dusty, not dirty" standard to eliminate extensive jobsite chemical flushes and reduce commissioning risk.

PROHIBITED CONDITIONS (Not Acceptable)

1. Construction debris
2. Burrs, slag, grinding residue, facing material
3. Fused connection flash or extrusions
4. Grease, oil, lubricants
5. Loose oxides, sugaring or corrosion
6. Adherent contamination not removable by flushing

REQUIRED CONDITIONS (Acceptable)

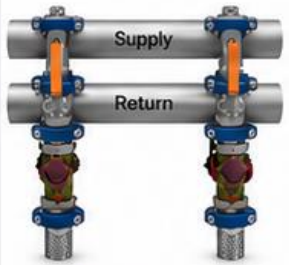
- ✓ Welds deburred and dressed
- ✓ Fused connections trimmed and cleaned internally
- ✓ Chemically passive internal surfaces
- ✓ No grease, oil, or hydrocarbons
- ✓ Particulate controlled to < 100 micron
- ✓ Residual particulate must be removable by flushing

THE RESULT:

- Faster Commissioning
- Less Water & Chemical Use
- Lower Labor & Cost
- Lower Risk Higher Reliability

EVERY COMPONENT. ONE STANDARD. CLEAN, DRY & INSTALLATION READY.

MANIFOLDS



Pre-cleaned, passivated inside and out

PIPING



Cleaned & passivated to remove scale, oxides & debris

BRANCH LINES



Pre-cleaned & passivated for clean connections

CDUs



Assembled clean with cleaned internal components

LOAD BANKS



Cleaned internal piping & heat exchangers

AIR SEPARATORS



Pre-cleaned vessel & internals

FILTRATION SYSTEMS



Clean housings, elements & internals

VALVES & FITTINGS



Cleaned, passivated & capped/sealed

Pre-cleaned. Pre-passivated. Particle-controlled < 100–200 micron. Because "dusty, not dirty" is the standard.

Compress Precommissioning: CDU Vs Flushing Rig

PREVIOUS GUIDANCE

"Thou shalt not use CDUs for flushing" – use dedicated flushing rigs instead.

KEY CONCERN:

PROJECT DELAY CAN COST

$300M/WEEK

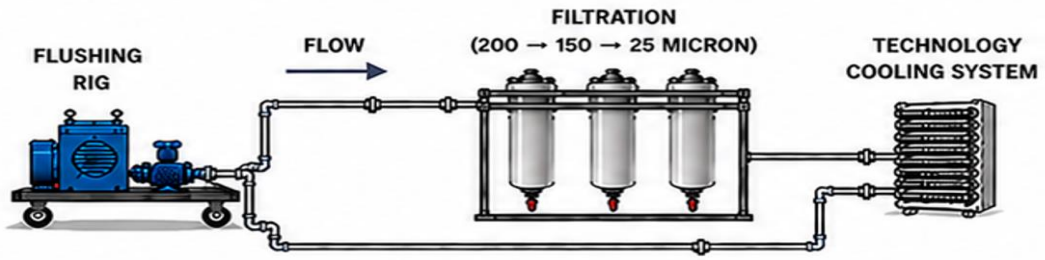
If using CDUs accelerates project completion, modifying procedures to support use of CDU for flushing makes sense.

EVOLVING POSITION

INLET FILTRATION PROTECTION
(Preferred Approach)

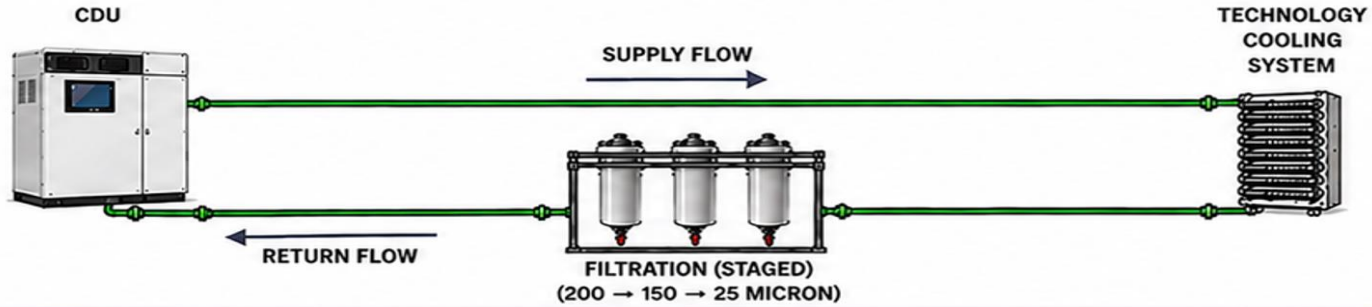
- Filtration (staged) at CDU inlet (200 → 150 → 25 micron)
- Protects CDU pumps and internals from debris
- Provides highest level of contamination control

DEDICATED FLUSHING RIG (TRADITIONAL APPROACH)



- Protects CDUs from debris and wear
- Often available when CDUs are not
- Designed for high-flow flushing
- Proven method with established procedures

CDU USED FOR FLUSHING



- Uses CDU pump and strainer(s) only
- No heat exchanger bypass needed
- Staged filtration 200 → 150 → 25 micron
- No harsh chemicals used
- Feasible per CDU manufacturers under conditions

KEY BENEFITS OF CDU FLUSHING

1. **Contamination control** — Effective multi-stage filtration removes debris that can impact performance and reliability.
2. **Cleaner commissioning baseline** — Achieves a cleaner system before CDUs arrive.
3. **Simpler transition to startup** — Less rework, fewer surprises, smoother commissioning.

Real-world hydraulics — Uses the same pumps and piping that will be used during operation.

WHEN FLUSHING RIGS SHOULD BE USED

Use a flushing rig when it saves time or does not impact cost and schedule.

CDU SUPPLY IS DELAYED
Flushing rigs are available to clean the system prior to the arrival of CDUs.

OR

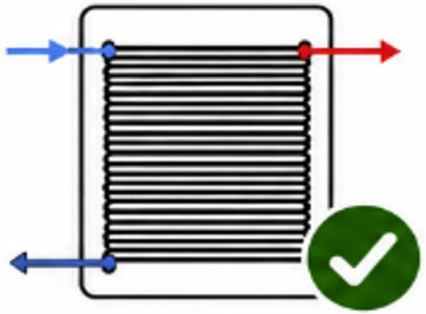
INADEQUATE FLUSHING FLOW
If CDUs cannot generate adequate flushing flow for the system.

Biostatic – Not Sterile

BIOSTATIC – THE RIGHT GOAL

Control microorganisms to safe levels.
Prevent biofilm from forming.

- Use PG25 + inhibitors + biocides
- Maintain chemistry within spec
- Monitor & test regularly
- Prevent biofilm from forming

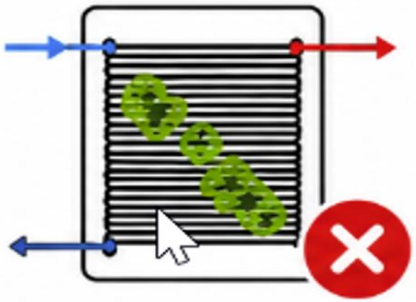


Heat exchangers cannot be sterile.
Focus on control, not elimination.

STERILE – NOT PRACTICAL, NOT NECESSARY

Sterility is impossible to maintain in heat exchangers.
Sterile pipes and connections provide no benefit.

- Sterility not achievable in HX
- No benefit for system reliability
- Adds cost & complexity
- Focus on what matters: chemistry control



Sterility is not achievable.
Chase chemistry, not sterility.

CHEMISTRY CONTROL IS KEY

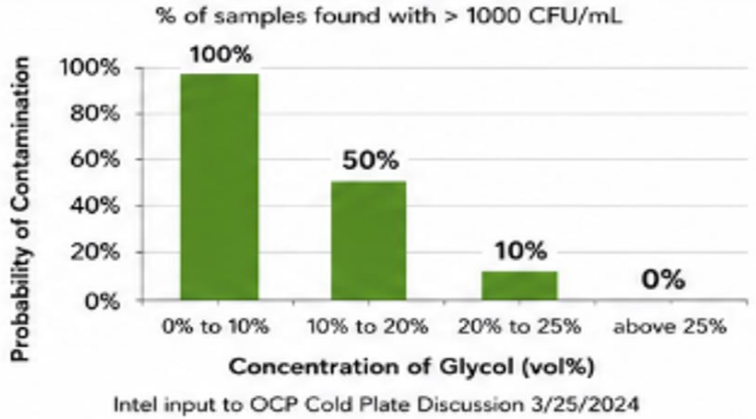
Microchannel cold plates and heat exchangers will foul if chemistry is not maintained.

OCP GUIDANCE

Follow OCP Cold Plate PG25 Guidance

Chloride < 25 ppm | Filtration ≤ 25 µm
Dissolved Oxygen Control | Staged Treatment

FOULING IMPACTS PERFORMANCE



Intel input to OCP Cold Plate Discussion 3/25/2024

POLISHED PIPES WON'T FIX BAD CHEMISTRY



Polished pipes
No impact on biofilm

Seamless pipes
No impact on biofilm

Sanitary connections
No impact on biofilm

The “Pipeline” from 80 KW/Rack to 1 MW/Rack

A community-driven roadmap for next-generation liquid cooling infrastructure

Community Driven | Liquid Cooling | Scalable Solutions | Open Innovation



From today’s deployments → Through evolving rack densities → To tomorrow’s 1 MW+ racks

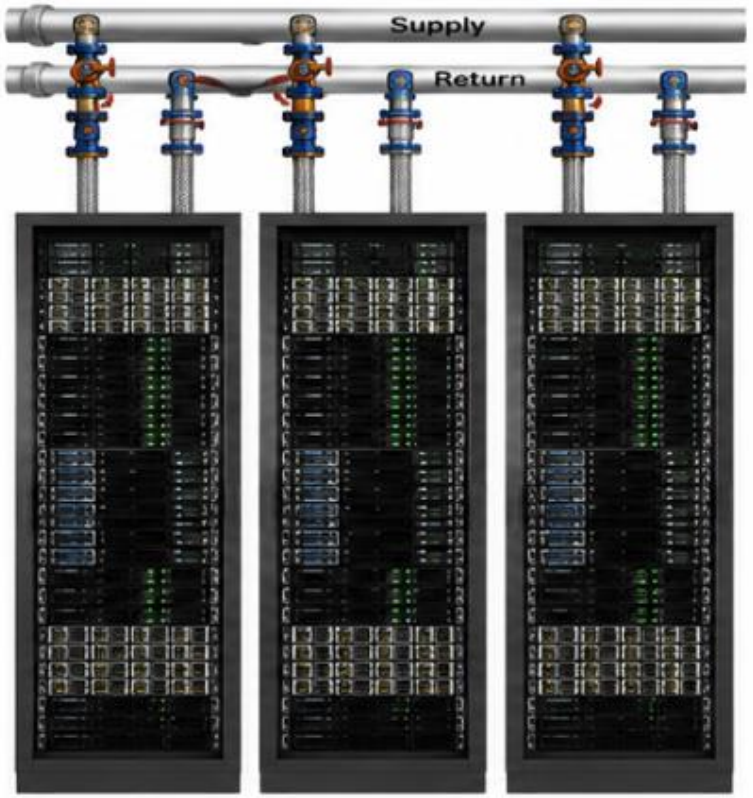
AI Rack Density Roadmap (2026-2029) As of Sept 2026

Vendor / Platform	2025 Current / Shipping	2026 Shipping / Production Ramp	2027 Announced	2028 Roadmap	2029+ Direction Only
NVIDIA	 **GB300 NVL72** **Up to 142 kW** ✓ Shipping	 **Vera Rubin NVL72** **Up to 230 kW** ✓ Production ramp (2H 2026)	 **Rubin Ultra** **(NVL72 / NVL144 / NVL576)** – Rack power TBD Announced	 **Kyber NVL144** **(144 GPU per rack)** – Rack power TBD Announced	 **Future NVIDIA Platform** **MW-class rack** **(> 600 kW per rack)** – Architecture and power TBD
AMD	 **MI350X** (64 GPU, air-cooled) – Rack power not publicly specified	 **MI355X** **Up to 140 kW** (128 GPU, liquid-cooled) ✓ Shipping (launched 2025)	 **Helios (MI455X)** **Up to 246 kW** (72 GPU, OCP ORW) ✓ Launching July 2026 (385 L/min)	 **Next-Gen AMD Platform** – Rack power TBD Announced (2027+)	 **Future AMD Platform** **600 kW – 1 MW+** – Architecture TBD Roadmap
Google	 **Ironwood (TPU7x)** 7th-generation TPU – Rack power not publicly specified ✓ Shipping	 **Next-Generation TPU** **(e.g., TPU v8)** – Rack power TBD Announced (Apr 2026)	—	—	—
Microsoft	—	 **Maia 200** AI accelerator (inference) – Rack power not publicly specified ✓ Shipping	—	—	—
aws	 **Trn3 UltraServer** **(up to 144 Trainium3 chips)** – Rack power not publicly specified ✓ Shipping	 **Trainium4** – Rack power TBD Announced	—	—	—

Data from Internet Searches,
Approximate values

Technology Cooling Lego© – 2”/DN50 Drops, 2”/DN50 Rack Connections

< 170 KW/Rack 6 fps (1.8 m/s)
< 230 KW/Rack 8 fps (2.4 m/s)



Pitch = 24” (600mm)

TCS Manifold Capacities (PG25, ΔT = 10°C, flow = 3 m/s)		
Manifold Size	Stainless Steel Schd 10	PPR/PPE/HDPE SDR 11
6”/DN150	~2500 KW	~1750 KW
8”/DN200	~4200 KW	~3000 KW
10”/DN250	~6600 KW	~4700 KW

< 340 KW/Rack 6 fps (1.8 m/s)
< 460 KW/Rack 8 fps (2.4 m/s)



Pitch = 48” (1200mm)

Liquid to Rack Pipe Sizing Tool

HEAT TRANSFER FLUID

Propylene Glycol 25%

Specific heat: 0.926 Btu/lbm·°F · Density: 8.609 lbm/US gal

PIPE SCHEDULE / MATERIAL

Stainless Steel Schedule 10

Determines wall thickness (and I.D.) for both lines at each nominal size.

Manifold & Branch Velocity / Pipe Size

Set the manifold (header) and branch (rack) velocity and pipe size independently. Typical practice targets roughly 10 ft/s (3 m/s) in the manifold and roughly 6 ft/s (1.8 m/s) in each branch — adjust to your project's design criteria.

Manifold / Header Line

VELOCITY (FT/S): 10 — 3.05 m/s

HEADER PIPE SIZE: 6" (DN 150)

Within typical manifold range (8–12 ft/s)

O.D. (IN)	WALL (IN)	I.D. (IN)	AREA (SQ IN)	FLOW (GPM)	FLOW (L/S)
6.625	0.134	6.357	31.739	989.3 (3,745.4 l/min)	62.42

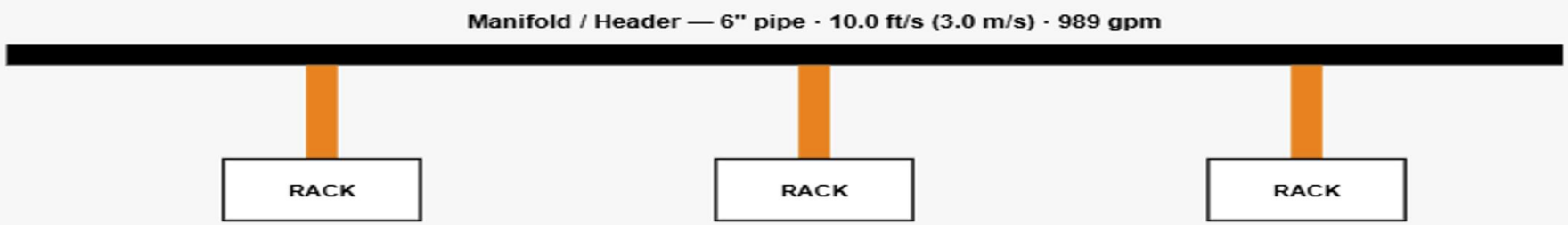
Branch / Rack Line

VELOCITY (FT/S): 6 — 1.83 m/s

BRANCH PIPE SIZE: 2" (DN 50)

Within typical branch range (4–8 ft/s)

O.D. (IN)	WALL (IN)	I.D. (IN)	AREA (SQ IN)	FLOW (GPM)	FLOW (L/S)
2.375	0.109	2.157	3.654	68.3 (258.7 l/min)	4.31



Illustrative schematic — number and spacing of branch lines is not to scale and does not reflect the rack count table below.

Cooling Capacity & Maximum Rack Count

For each ΔT, **Max Row kW** is the header/manifold pipe's carrying capacity at the manifold velocity above, and **Max Rack kW** is a single branch's carrying capacity at the branch velocity above. A rack-count column shows the number of racks a row can serve at that kW/rack, or a dash if a single branch cannot even deliver that much to one rack.

`kW = Mass flow rate (kg/s) × Specific heat (kJ/kg·°C) × ΔT (°C)`

Reset to defaults

ΔT (°F)	ΔT (°C)	MAX ROW kW (MANIFOLD)	MAX RACK kW (BRANCH)	RACKS @ 100 kW	RACKS @ 135 kW	RACKS @ 160 kW	RACKS @ 220 kW
18	10.0	2,496.2	172.4	24	18	15	—

TCS Pipe Sizing Tool - Size & flow vs KW

ITE Rack Piping KW Decision Tree – Branch Line flow 8 fps (m/s)

Is the rack heat load ≤ 230 KW?

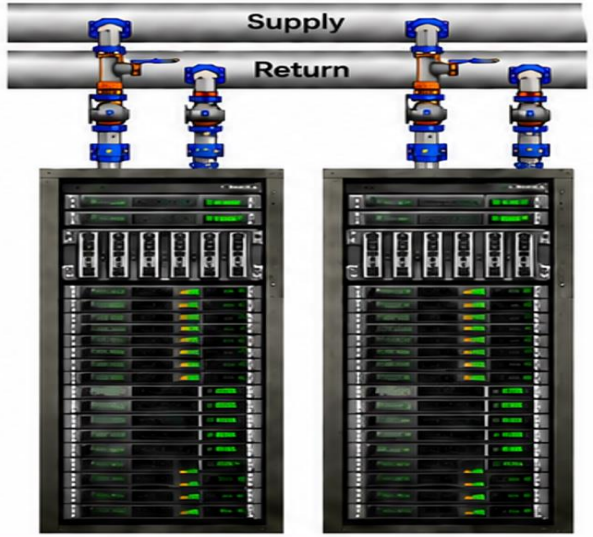
YES (≤ 230 KW)

NO (> 230 KW)

OPTION 1

Up to 230KW/ Rack (24”/600mm pitch) – 2”/DN 50

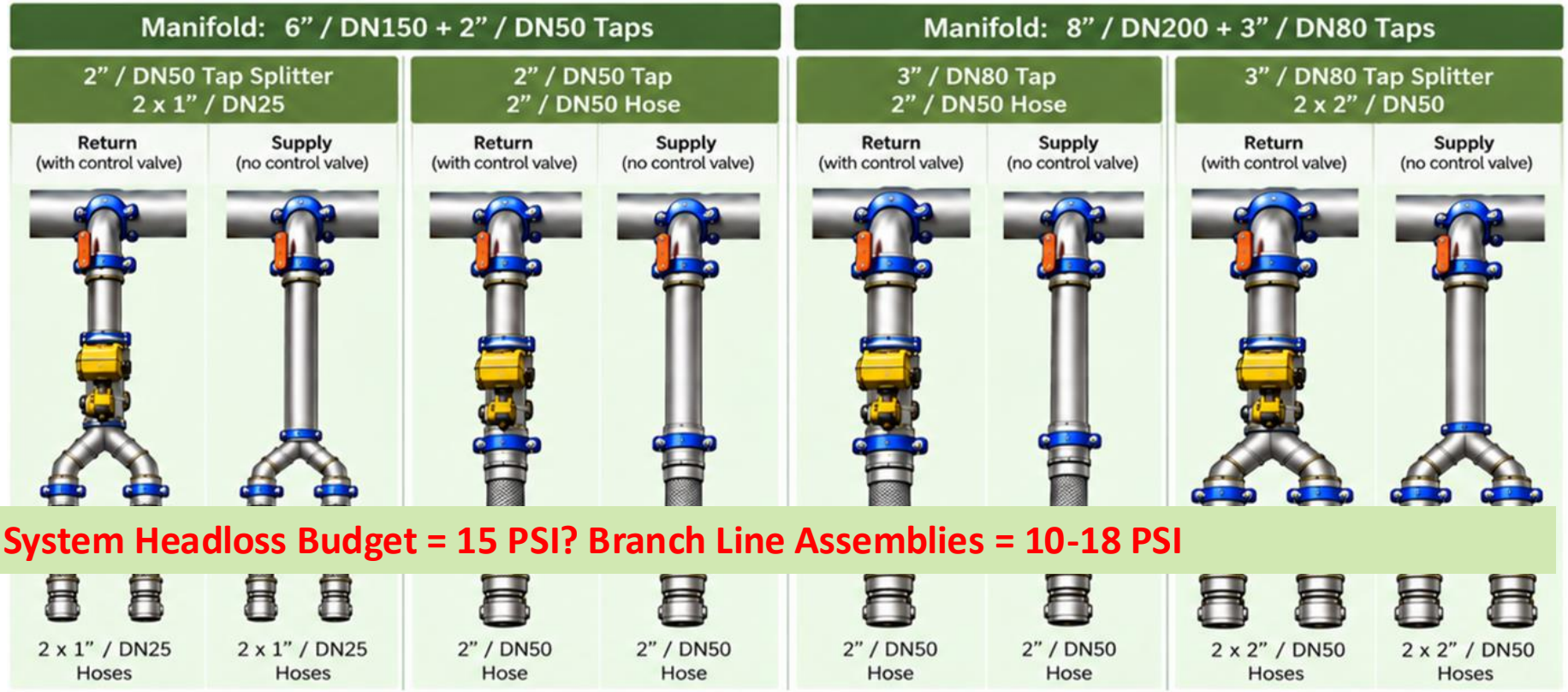
For 24” (600mm) racks up to 230KW use something similar to this image:



2”/DN50 rack connections (24”/600mm pitch)

Reference Designs – Valve Trains (Return & Supply)

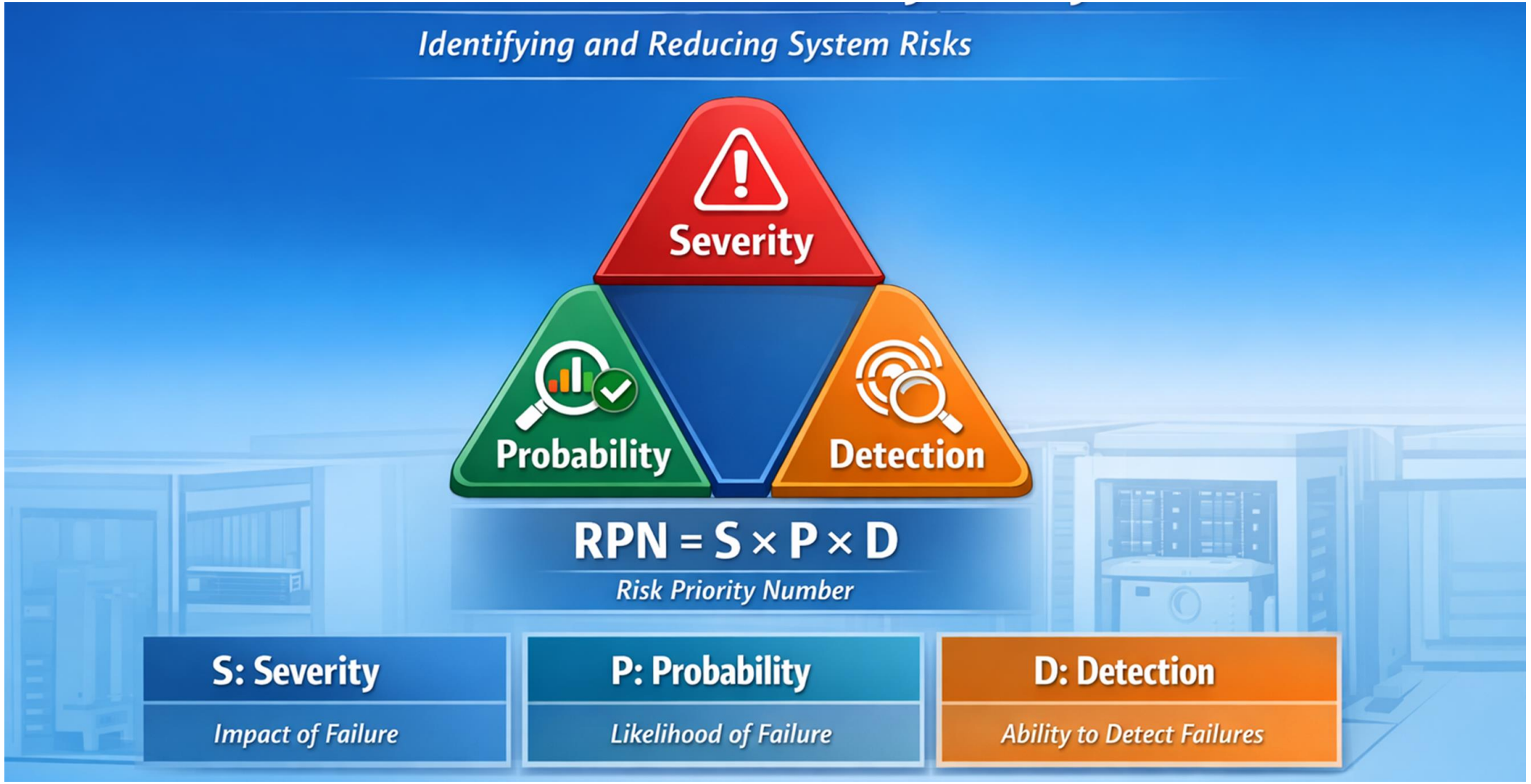
Loss point	Range (psi)
Branch tee + isolation valve (3")	0.2 – 0.6
3" drop friction (10–15 ft + fittings)	0.3 – 0.8
Control valve	See Section 4
3" → 2" split / reducing wye	0.2 – 0.5
2" flexible hose (6–10 ft)	0.4 – 1.5
Rack isolation + quick disconnects	0.5 – 2.0



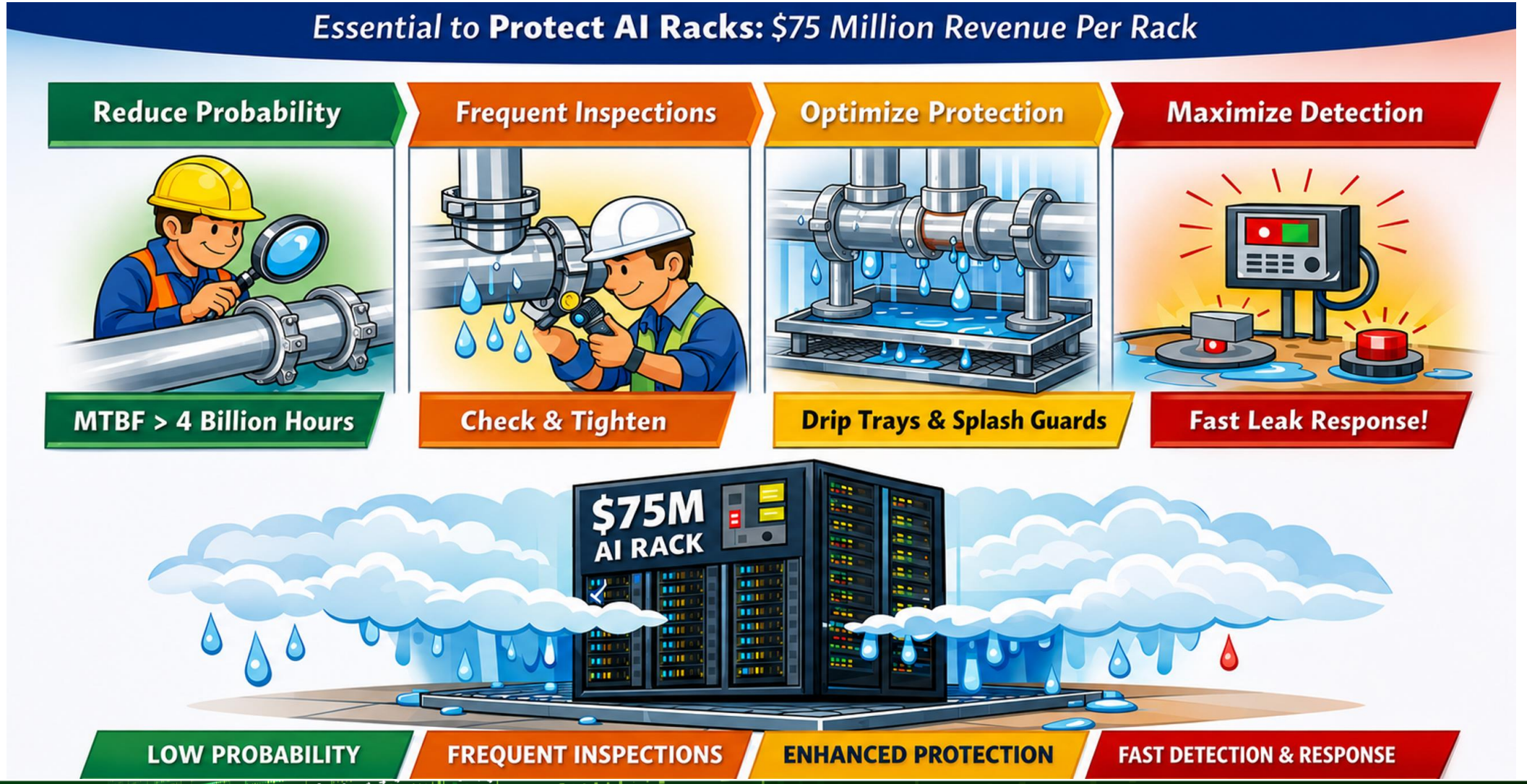
System Headloss Budget = 15 PSI? Branch Line Assemblies = 10-18 PSI

Parameter	6" / DN150 Header with 2" / DN50 Tap		8" / DN200 Header with 3" / DN80 Tap	
Header Capacity @10 fps (3 m/s)	2.5 MW		4.3 MW	
Splitter	2" / DN50 – 2 x 1" / DN25	NA	NA	3" / DN80 – 2 x 2" / DN50
Rack Connection	1" / DN25	2" / DN50	2" / DN50	2" / DN50
Capacity @6 fps (1.8 m/s)	2 x 45 = 90 kW	172 kW	172 kW	2 x 172 = 344 kW
Capacity @8 fps (2.4 m/s)	NA	230 kW	230 kW	2 x 230 = 460 kW

FMEA for Tech Cooling Reliability Analysis



Preventing Leaks in Tech Cooling Systems



Mean Time Between Failure (MTBF) vs Repair (MTBR)

MTBF – Mean Time Between Failures

How often does a connection unexpectedly leak?

$$\text{MTBF} = \frac{\text{Total Connection-Hours of Operation}}{\text{Number of Leakage Failures}}$$

Large field populations with no failures provide a statistical lower bound on MTBF.

Example: >150,000 couplings, 18 data center projects, 15 years of operation, no leakage failures → Actual MTBF > 9 Billion hours*

**Lower confidence bound, not zero risk.*

MTBR – Mean Time Between Repairs (Planned + Unplanned)

How often must the connection be opened or serviced?

$$\text{MTBR} = \frac{\text{Total Uptime (Connection-Hours)}}{\text{Total Number of Repairs (Planned + Unplanned)}}$$

Maintenance requirements drive lifecycle exposure.

Example: 100 MW TCS with ~30,000 connections and MTBR of 2 years → ~15,000 interventions per year.

Each intervention introduces risk of disassembly, reassembly error, contamination, air introduction, and recommissioning.

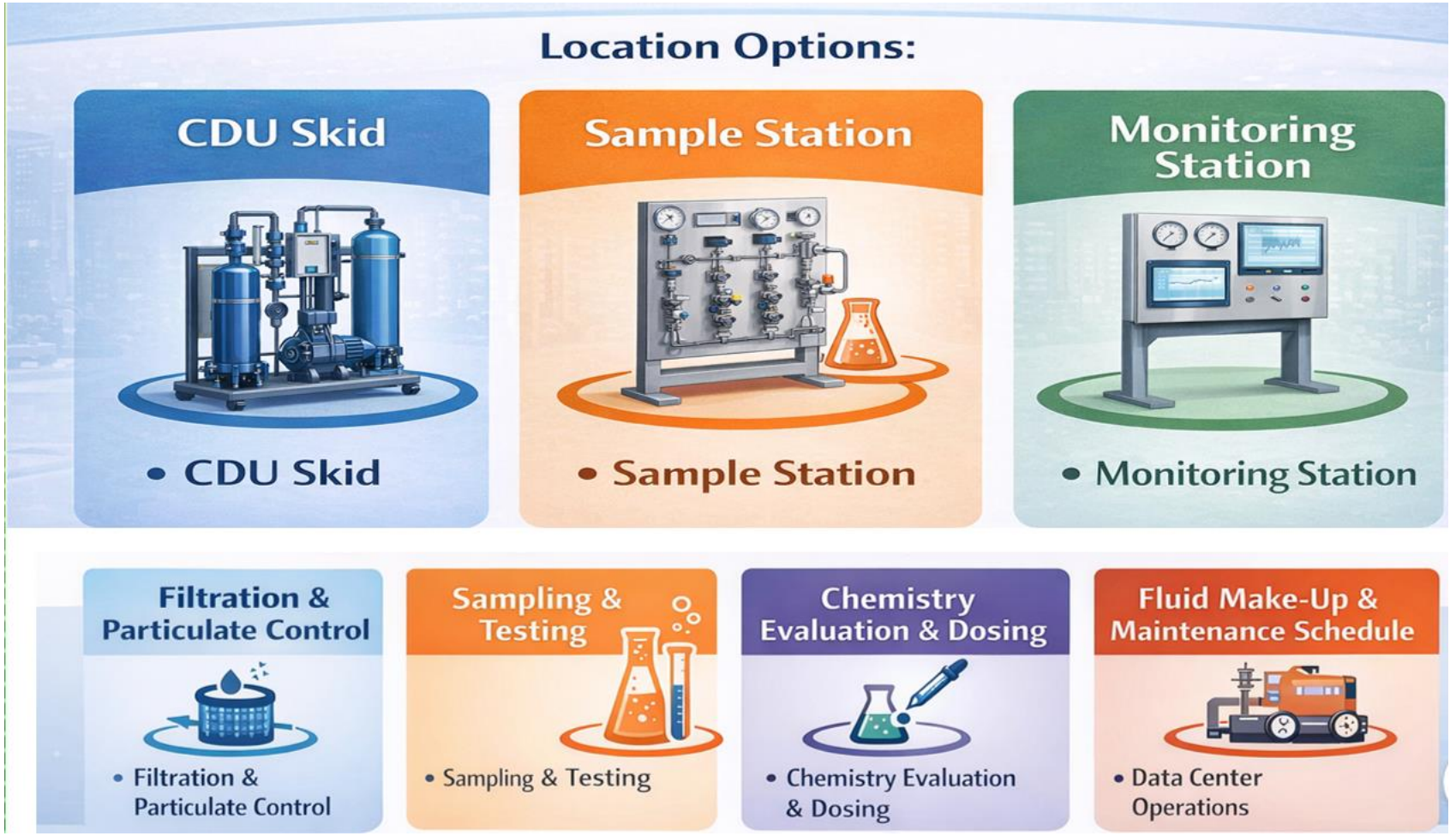
Why Both Metrics Matter

MTBF addresses the probability of unexpected leakage failure.

MTBR addresses how often the connection must be opened or serviced.

Both influence the FMEA probability of occurrence over the system lifecycle.

Technology Fluid Control & Maintenance



Call to Action – Get Ready for AI Factories!

Technology Cooling Workstream Leaders

- Don Mitchell – Victaulic
- Hussam Kabbani – Google

OCP "Ready for AI"

- Madhu Iyengar - Google

DCF Project Leaders

- Saket Karajgikar – NVIDIA
- Vivek Prasad – Microsoft
- Scott Sharp – Google

Modular TCS workstream

Modular TCS WP Rev 2

Modular TCS Webpage

Modular TCS Webpage & Wiki

Monthly Call

TCS Pipe Sizing Tool

TCS Pipe Sizing Tool - Size & flow vs KW

Open Discussion

